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# Acute toxicity of zinc with respect to $LC_{50}$, morphological changes and behavioural responses in freshwater fish *Clarias batrachus*

**Kritika Pandey*, Farha Ashique, Dileep Kumar & Parimal Kumar Khan**

Department of Zoology, Patna University, Patna, Bihar, India

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**Abstract-** Zinc is an essential heavy metal that plays a crucial role in various biological processes. Its excessive presence in aquatic system, however, possess substantial risks to aquatic life. This study aimed to determine the acute toxicity of zinc sulphate ($ZnSO_4$) in terms of median lethal concentrations ($LC_{50}$), as well as morphological and behavioural changes in the catfish *Clarias batrachus* upon its short-term exposure. Physico-chemical parameters of the water, such as pH, temperature, dissolved oxygen, total hardness, and total alkalinity, were consistently monitored and maintained within optimal ranges which include 176±3 mg/l $CaCO_3$ hardness and pH 7.0±0.5. The results revealed a dose-dependent increase in mortality rates as well as marked behavioural and morphological changes. The $LC_{50}$ value of zinc sulphate was determined as 170 mg/l through probit analysis. Severely altered morphological and behavioural patterns were observed upon exposure to zinc at its acute levels. Behavioural responses included loss of equilibrium, hyperactivity, loss of reflexes, gasping at the surface, increased mucus secretion, and morphological effects were discoloration, swollen abdomen, posturing of pectoral fins and occasional haemorrhaging. Symptoms were more prominent when fish were exposed to higher levels of zinc sulphate, indicating induction of physiological stress upon exposure to excessive zinc. The observations highlighted the ecological risks posed by zinc contamination in aquatic systems signifying the need of developing effective environmental protection strategies.

**Keywords: *Clarias batrachus*, zinc sulphate, $LC_{50}$, morphological effects, behavioural changes**

## INTRODUCTION

Zinc is a trace element required in biological organisms to act as a cofactor for numerous enzymes involved in vital biochemical processes of cellular metabolism.[1] It is indispensable for immunity, wound healing and cell division.[2] Despite its critical biological roles, concentration of zinc above its threshold in aquatic ecosystem, largely due to anthropogenic activities like industrial discharges, mining operations, and agricultural runoff, appears to be harmful to aquatic organisms.[3] Zinc pollution in aquatic systems is a pressing environmental issue, exacerbated by its non-biodegradable nature, and accumulation in the sediments and biota.[4] The bioavailability and toxicity of zinc is however influenced by various factors, including water hardness, pH and temperature.[5]

*Clarias batrachus*, the walking catfish, is a freshwater species distributed across Southeast Asia and parts of South Asia.[6] This species is known for its tolerance to environmental fluctuation, including low oxygen levels and changing water temperatures, which make it an ideal model organism for ecotoxicological studies. It has been widely used to monitor environmental pollution including toxicity of zinc, owing to its ecological relevance, ease of maintenance in laboratory conditions, and sensitivity to

*Corresponding author :
Phone : 8005430349
E-mail : kritika1pandey@gmail.com

various contaminants.[7] Das and Das (2012)[8] investigated the effects of zinc sulfate on *C. batrachus* and reported an $LC_{50}$ value (1.7 g/L) over an exposure period of 96-hour. Later, Gupta *et al.* (2009)[9] determined a lower $LC_{50}$ value (4.1 mg/L) for *C. batrachus* exposed to zinc sulfate for 96 hours. The $LC_{50}$ value of zinc obtained through sigmoid curve in *C. batrachus* is 17.22 mg/l.[10] The $LC_{50}$ value of $ZnSO_4$ following in acute toxicity was however found to be 3.1, 2.53, 2.03 and 1.75g/l at 24, 48, 72, and 96 hrs respectively.[8] Variations in $LC_{50}$ values reported in these studies can be attributed to differences in experimental conditions, such as water hardness, pH, and temperature, as well as the physiological state of the fish.

Behavioral responses and morphological changes in *C. batrachus* to zinc sulfate exposure have also been studied. Behavioral changes often occur rapidly in response to toxicants, and can provide early warning signals of environmental contamination.[11] Reed *et al.* (1980)[12] observed several behavioral changes in the fish exposed to zinc sulfate which include increased hyperactivity, erratic swimming and elevated gill ventilation. These responses are indicative of physiological perturbations which can impair the ability of fish to forage, evade predators and reproduce. In addition to behavioral changes, zinc exposure may lead to significant morphological alterations in fish. Mishra and Mohanty (2008)[13] reported various morphological changes in *C. batrachus* exposed to zinc sulfate, including gill hyperplasia, liver necrosis and kidney lesions. These findings reflect occurrence of severe physiological damage, and highlight the importance of assessing both behavioral and morphological impacts to understand the extent of zinc toxicity. The induced behavioural and morphological changes, therefore, underscore the ecological impact of zinc pollution.

## MATERIAL & METHODS

Healthy specimens of *C. batrachus* (Figure 1), measuring 10-12 cm in length and 18-22 g in weight, were obtained from local fish farm at Patna, Bihar (India). It belongs to class actinopterygii, order siluriformes and family clariidae. Specimens of this species were initially treated with 0.5 % $KMnO_4$ for five minutes for dermal disinfection followed by their acclimatisation for period of 15 days in laboratory condition, and were feed on small pieces of earthworm. The Zinc sulphate ($ZnSO_4.7H_2O$) was used for the assessment of heavy metal exposure to fish specimens. Acclimated fish specimens were kept starved for 24 hours before the start of the test. The control group specimens were not subjected to zinc sulphate exposure.



**Figure 1: Experimental fish specimen *Clarias batrachus***

The physicochemical parameters of the tap water were noted periodically as per APHA (2017)[14] before the determination of 96 h $LC_{50}$ (Table.1). Upon exposure of the fish to various concentrations of the zinc sulphate, observations were made on the behavioral and morphological responses of the fish at 24, 48, 72, and 96 hours. Control fish specimens were monitored simultaneously to provide a reference for assessing the occurrence of any behavioral or morphological changes. The behavioral and morphological indicators observed include loss of equilibrium, general activity, startle response, haemorrhage and deformity. Startle responses were monitored by the following procedures in sequence: passing hand over the test tank (overhead moving visual stimulus), rapping on the tank (vibration stimulus), and lightly touching the fish with a stick (tactile stimulus).

## RESULTS

The relationship between the different exposure concentrations of zinc sulphate and the mortality response rate of *C. batrachus* was observed at selected physicochemical characteristics (Table 1) maintained in the aquaria during toxicity testing of the given fish species. The mortality rate in the control group did not exceed 10% and the remaining specimens of fish looked healthy throughout the experiment. Mortality of fish was observed at various levels of zinc exposure during the estimation of $LC_{50}$ (Table 2).

**Table 1: Physicochemical parameters of water recorded and maintained during the experimentation.**

| Sl. No. | Physicochemical parameters | Value |
|---|---|---|
| (i) | pH | 7.0±0.5 |
| (ii) | Temperature | 25°C±2°C |
| (iii) | Dissolve oxygen | 6.5±0.3 mg/L |
| (iv) | Total Hardness | 176±3 mg/L |
| (v) | Total alkalinity | 74±3 mg/L |

**Table 2: Showing dose- responses during the determination of $LC_{50}$ in *Clarias batrachus* at a 96-hour exposure period.**

| Groups | Dose (mg/L) | Log value | Mortality (%) | Corrected mortality (%) | Probit value |
|---|---|---|---|---|---|
| 1 | 120 | 2.08 | 0 | 2.5 | 3.04 |
| 2 | 150 | 2.18 | 20 | 20 | 4.16 |
| 3 | 160 | 2.20 | 30 | 30 | 4.48 |
| 4 | 190 | 2.28 | 70 | 70 | 5.52 |
| 5 | 200 | 2.30 | 80 | 80 | 5.84 |
| 6 | 220 | 2.34 | 100 | 97.5 | 6.96 |

Further, the probit line graph of acute toxicity of zinc on *C. batrachus* was drawn based on values mentioned in table-2, leading to calculation of $LC_{50}$ of zinc sulphate. The $LC_{50}$ value was calculated by the method of Miller and Tainter (1944), as mentioned by Randhawa (2009)[15], and it was found 170 mg/l at a 96-hour exposure period.

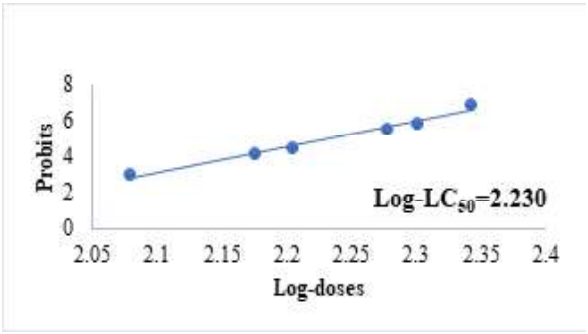



**Figure 2: Plot of log-concentration of zinc sulfate versus probit values.**

Fish specimens while exposed to varying concentrations of zinc sulfate exhibited the gradual progression in behavioral responses and morphological changes which were more prominent with increase in exposure level (Table 3).

**Table 3: Behavioural and morphological effects of zinc sulphate *C. batrachus* (Linn.)**

| Dose (mg/L) | Behavioural changes | Morphological changes |
|---|---|---|
| Upto 50 | No visible changes | No visible changes |
| Upto 100 | Startle response, hyperactive, increased swimming rate, tendency to jump out of the tank | Slight discoloration |
| Upto 170 | Hypoactive, erratic swimming, frequent surfacing | Mild discoloration, body appeared covered with thin white film |
| Upto 190 | Significant reduction in movement, loss of equilibrium | Severe discoloration, pronounced fin erosion |
| Upto 250 | Immobile, loss of reflexes, gasping at the surface | Extensive haemorrhaging, fin erosion, bloated abdomen |

## DISCUSSION

Fishes constitute a valuable commodity from the stand point of human consumption; hence, accumulation of heavy metals in the body of aquatic biota including fishes becomes a serious concern from human health point of view. Upon exposed to elevated level of metals in aquatic ecosystem, these organisms tend to take metals up from their direct environment.[16] Heavy metal contamination of water bodies may also have adverse effects on ecological balance of recipient environment as well as on the survival and diversity of aquatic organisms.[17] Some studies were carried out to assess the toxicity of zinc sulphate in fishes (Das and Das 2012[8]; Joshi 2011[17]; Gul *et al.* 2009[18]). Exposure to Zinc in fishes can result in damage of their organs like gills, liver and kidneys, leading to various behavioral, physiological and biochemical changes.[19]

In the present study, the $LC_{50}$ value of zinc at 96 hours in the given fish was found relatively high (170mg/l) compared to the values reported in some other studies in the *Clarias* (Table 4). Higher value of $LC_{50}$ observed in this study might be due to several factors which particularly include physicochemical parameters of water, and use of young and juvenile fish specimens in the present study which are supposed to have greater tolerance to zinc toxicity compared to specimens of higher age group. Das and Das (2012)[8] have also reported even greater $LC_{50}$ value (1.75 g/l) of $ZnSO_4$ in *C. batrachus*.

Behavioral changes in fish are often the earliest manifestations of toxic exposure, and can provide crucial insights into the sub-lethal effects of pollutants. In this study, *C. batrachus* exposed to zinc sulfate exhibited marked behavioral alterations such as initial immobility, rapid and erratic swimming, restlessness, and increased surfacing tendency. These observations are consistent with previous findings by Joshi (2011)[17], attributing such behaviour responses to the neurotoxic effects of the metal. The initial period of immobility observed in this study could therefore be a stress response to the sudden exposure to the toxic substance. Sprague (1969)[20] have noted that such behavior often precedes more active avoidance responses in fish exposed to heavy metals. As the exposure continued, the fish exhibited rapid and erratic swimming behavior that can be linked to attempts of fish to escape the toxic environment. Giattina *et al.* (1982)[21] also observed similar avoidance behaviors in fish exposed to copper and nickel. This increased swimming activity appears to be a survival

mechanism, and results in higher energy expenditure which can further exacerbate the stress experienced by the fish. Drummond *et al.* (1986)[22] reported that such behaviors, when prolonged, lead to significant energy depletion, weakening of the fish and making them more vulnerable to other environmental stressors.

In addition to behavioral changes, morphological alterations were also observed in *C. batrachus* following exposure to zinc sulfate. The fish developed signs such as fin erosion, skin discoloration and excessive mucus secretion. These morphological changes are in agreement with the findings of Svecevieus (2001)[23], highlighting that the damage to the integumentary system serves as the first line of defense against environmental toxins. Further studies have shown that zinc mediated morphological and behavioral changes have strong corelation with the induced alterations in hematological parameters in *C. batrachus*, indicating that zinc affects not only external features but also internal physiological processes which disrupt normal functions and increase mortality rate.[24] The observed fin erosion and skin discolorations are of particular concern, as breakdown of protective barrier may lead to increased vulnerability to other pollutants and pathogens. The erratic swimming and loss of equilibrium observed in this study point to the neurotoxic effects of zinc.

Thus, the exposure of *C. batrachus* to zinc sulfate has resulted in significant behavioral and morphological changes, reflecting the toxic impact of zinc on physiological functions of the fish including on central nervous system. Hence, considering the significant risks posed by zinc contamination, there is an urgent need for stringent regulation and monitoring of zinc levels in freshwater environments to protect aquatic life and maintain ecological balance.

## CONCLUSION

This study highlights the acute toxicity of zinc sulfate on the freshwater fish *C. batrachus*, revealing observable impacts on both behavior and morphology. The experiment to calculate $LC_{50}$ value demonstrated that increase in mortality occurred in a dose-dependent manner underscoring the toxic potential of zinc. Behavioral changes were induced at higher level zinc exposure showing signs of distress such as hyperactivity, erratic swimming, and attempts to escape. It was followed by marked reductions in movement, loss of balance, and severe respiratory distress which were accompanied by significant morphological changes such as discoloration, fin erosion, and haemorrhaging. It suggests the occurrence of stress and potential neurological effects caused by zinc exposure. This may lead to broader ecological consequences, including disturbances in population dynamics and trophic levels within the ecosystem. Given zinc's pervasive presence in municipal waste and its potential for ecological harm, it is crucial to monitor the presence of zinc in aquatic environments. The findings from this study reinforce the need for further research on long-term and chronic effect of zinc on aquatic organism including fishes.

**Table 4: Showing the variation in 96h-$LC_{50}$ value of zinc observed by various researchers**

| Species | 96-h $LC_{50}$ value | References |
|---|---|---|
| *Clarias batrachus* | 7.50mg/l | Kumar and Kumar (2013)[25] |
| *Clarias batrachus* | 8.21 ppm | Kumar *et al.* (2015)[26] |
| *Clarias batrachus* | 17. 22 mg/l | Joshi (2011)[17] |
| *Clarias batrachus* | 1.75 g/l | Das and Das (2012)[8] |
| *Clarias batrachus* | 37.22 mg/l | Srivastava (2019)[27] |
| *Clarias gariepinus* | 36.7 mg/l | Ololade and Ogini (2009)[28] |
| *Clarias gariepinus* | 65.151 mg/l | Chidiebere (2019)[29] |
| *Clarias gariepinus* | 78.178 mg/l | Ezeonyejiaku *et al.* (2010)[30] |

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